

CSCE 649 PHYSICALLY-BASED MODELING · MODULE 10 · LECTURE NOTES

Cloth and Hair

Thin materials, where everything is harder.

MODULE OUTCOMES

- Explain why thin materials are numerically difficult.
- Implement implicit integration for cloth.
- Apply strain limiting and explain what it costs.
- Handle cloth self-collision robustly.
- Explain why hair needs different methods from cloth.

1 Why thin materials are hard

STIFF IN ONE DIRECTION, FLOPPY IN ANOTHER

Stretching: woven fabric extends by a few percent at most before the threads themselves must stretch. Very stiff in-plane.

Bending: the same fabric folds almost freely — the resistance to bending is smaller than the resistance to stretching by several orders of magnitude.

That ratio is exactly the numerical definition of stiffness (Module 02 §3): timescales differing by orders of magnitude within one system.

So cloth is stiff by construction, not by parameter choice. An explicit integrator must resolve the fast in-plane modes, where nothing visually interesting happens, while all the visible motion — draping, swinging, wrinkling — occurs in the slow floppy direction. There is no setting that avoids this.

- **Large surface area for its mass.** A square metre of fabric has vastly more opportunity for contact than a boulder of the same weight, and contact is the expensive part.
- **Self-collision is everywhere.** Every triangle can contact every other triangle of the same mesh, minus the topologically adjacent ones (Module 05 §5).
- **Once tangled, cloth does not recover.** There is no local information indicating which way to separate two interpenetrating layers, and pushing apart arbitrarily usually worsens the tangle.
- **Contact is the visible phenomenon.** Drape, folds, and wrinkles are all consequences of cloth contacting itself and other objects — so contact cannot be approximated away as it often can for rigid bodies.
- **The combination is what makes cloth hard:** it requires the most difficult integration and the most difficult collision handling simultaneously, and the two interact.

2 Implicit cloth

BARAFF AND WITKIN, 1998

Before this paper, cloth simulation required thousands of tiny explicit time steps per rendered frame, because the stretch stiffness bounded the step size by Module 02's $\Delta t < 2\sqrt{(m/k)}$.

Baraff and Witkin applied backward Euler and took a single step per frame. Unconditional stability meant the step size was limited by what looked right rather than by the stiffest spring in the mesh — a change of several orders of magnitude in cost.

The price is a large sparse linear system per step, solved by conjugate gradient, and the substantial artificial damping inherent to backward Euler (Module 02 §3). Cloth simulated this way can look over-damped, losing the high-frequency flutter that reads as fabric.

They also introduced filtered constraints: a way to impose constraints — pinned vertices, contacts — by filtering the search directions inside the CG iteration, so no separate constraint solver is needed. It is an elegant trick and is why the paper is still read.

Component	Detail
Force Jacobian $\partial f/\partial x$	Derived per element and assembled. Sparse and symmetric. Deriving it correctly for bending forces is the tedious part of the implementation.
Conjugate gradient solve	The dominant per-step cost. CG requires only matrix-vector products, so the Jacobian never needs to be assembled explicitly — compute Jv directly by looping over elements. This is a large saving in memory and time.
Preconditioner	A diagonal (Jacobi) preconditioner costs almost nothing and substantially reduces iteration count. Do it.
Filtered constraints	Pinning and contact imposed by projecting the CG search directions.
Warm starting	Use the previous step's solution as the initial guess. The answer changes little between frames, so this reduces iterations considerably.

3 Strain limiting

LIMIT THE STRAIN RATHER THAN RESISTING IT

The stiffness dilemma from Module 03: cloth must barely stretch, which requires a very stiff spring, which requires a tiny time step.

Strain limiting sidesteps it. Use a spring of moderate stiffness, and then, as a separate pass, **directly clamp any edge that has stretched beyond a threshold** — Provot's original proposal used 10% — by moving its endpoints toward each other.

This is a constraint projection (Module 04), not a force. It costs nothing in stability, because moving a point to a valid position cannot destabilise anything.

The costs are real. It is not derived from any physical principle; it removes energy in an uncontrolled way; and **it can fight other constraints** — a strain-limiting pass can undo contact resolution performed immediately before it, so ordering and iteration between the passes matters.

3.1 Bending

- **A spring between vertices two apart** (Module 03) is the crude approach. It does resist folding, and it also resists stretching along the same direction, so the two behaviours are coupled and cannot be tuned independently.
- **Dihedral angle bending** measures the angle between adjacent triangles across their shared edge and applies a restoring torque. This is the correct formulation and decouples bending from stretching; the derivatives needed for an implicit solve are involved but standard.
- **Real fabric is anisotropic in bending.** Woven cloth folds more readily along the grain than across it, and more readily still on the bias. This is specifiable in a continuum model and not in a spring model.
- **And fabric has bending hysteresis:** a crease stays creased. Representing that requires plasticity in the bending model — a rest angle that updates when the bend exceeds a threshold.
- **Most production systems ignore both** and produce acceptable results, which is worth knowing before spending a week on it.

4 Robust collision

Bridson, Fedkiw and Anderson's 2002 treatment is the standard, and its structure is more instructive than any individual component: **it is layered, and the final layer cannot fail.**

- 1 **Give the cloth a thickness.** Treat each triangle as having a small offset volume and keep surfaces separated by that distance, rather than detecting overlap after it has occurred. Most potential problems never become problems.
- 2 **Repulsion forces.** Gentle forces applied early, as surfaces approach within the thickness. These resolve the large majority of contacts smoothly, before they become hard constraints.

- 3 **Continuous collision detection** for whatever the repulsions did not catch — both vertex–triangle and edge–edge, with exact time-of-impact computation. Expensive, and applied to a much smaller set because of steps 1 and 2.
- 4 **Rigid impact zones as a last resort.** If a region is so tangled that collision resolution will not converge, **freeze the whole region into a single rigid body for the remainder of the step.** Completely unphysical, visible if it happens often — and it **guarantees termination**, which nothing else in the list does.

The layering is the method. Each stage handles what the previous missed, at increasing cost, and the final stage cannot fail. This pattern — a cheap common case backed by an expensive guaranteed fallback — recurs throughout production simulation and is worth adopting generally.

5 Hair

HAIR IS NOT THIN CLOTH

A strand is one-dimensional, so it has bending *and* twisting degrees of freedom, both of which matter for how it looks. Cloth largely ignores twist; a hair strand's twist determines whether a curl holds its shape.

There are on the order of 100,000 of them on a human head. Pairwise strand–strand collision among 10^5 strands is not feasible, and that is before contact with the head and body.

Strand–strand friction dominates the appearance. Hair holds a shape not because individual strands are stiff — they are not — but because strands grip each other. A simulation with correct strand mechanics and no inter-strand friction looks like wet hair, always.

So hair is simulated as a few hundred guide strands with the rest interpolated, and friction modelled as a continuum effect — a density-dependent drag — rather than pairwise. **Cosserat rods** are the accurate formulation for a single strand, handling bending and twisting correctly, and are used where individual strands matter.

Technique	Buys	Costs
Implicit integration	Large stable time steps.	A sparse linear solve per step, and heavy artificial damping.
Strain limiting	Inextensibility without stiffness.	Unphysical; removes energy uncontrollably; fights other constraints.
XPBD (Module 04)	Unconditional stability with a meaningful stiffness parameter.	Not derived from mechanics; material behaviour is approximate.
Cloth thickness and repulsion	Robust collision; most contacts never become hard.	Additional detection work every step.
Rigid impact zones	Guaranteed termination.	Visibly frozen regions when they trigger often.

Every row is a trade, and none of them is free. That is the honest summary of cloth simulation as it currently stands: a collection of techniques each solving one problem and introducing another, assembled

into a pipeline that works well enough. There is no formulation that is simultaneously stable, fast, physically correct, and robust under contact, and anyone claiming otherwise has not tried it on a difficult shot.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Baraff & Witkin — Large Steps in Cloth Simulation (1998, free)	The implicit cloth paper of §2, including filtered constraints. Still the standard reference.
Bridson, Fedkiw & Anderson — Robust Treatment of Collisions, Contact and Friction for Cloth Animation (2002, free)	The layered collision treatment of §4, including rigid impact zones. Essential reading.
Provot — Deformation Constraints in a Mass-Spring Model (free)	The original strain limiting of §3.
Bergou et al. — Discrete Elastic Rods (free)	The standard formulation for hair and strands in §5, handling bending and twisting correctly.
Ten Minute Physics — cloth episodes	XPBD cloth with working code — the practical alternative to implicit integration.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 Measure the stiffness ratio between stretching and bending for your cloth model and relate it to the stability limit from Module 02.
- 2 Implement implicit (backward Euler) cloth with conjugate gradient. Report the step size now achievable against the explicit limit.
- 3 Implement the matrix-free matrix-vector product and compare memory use and speed against assembling the Jacobian.
- 4 Add a diagonal preconditioner and report the reduction in CG iterations.
- 5 Measure the energy backward Euler removes at several step sizes, and describe how the cloth's appearance changes.
- 6 Implement strain limiting. Compare the maximum stretch and the maximum stable step against a stiff-spring approach.

- 7 Show strain limiting fighting contact: construct a case where the limiting pass pushes a vertex back into a collider, then fix it by iterating the two passes.
- 8 Implement dihedral angle bending and compare against the two-apart-spring approach on a draping test.
- 9 Implement cloth thickness and repulsion forces. Report what fraction of contacts are handled by repulsion alone.
- 10 Implement a rigid impact zone fallback and construct a tangle that triggers it. Document how visible the freeze is.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is cloth numerically stiff by construction rather than by parameter choice?
- 2 Give four reasons cloth collision is harder than rigid body collision.
- 3 What did Baraff and Witkin change, and what did it cost?
- 4 Why does conjugate gradient not require the Jacobian to be assembled?
- 5 What is strain limiting, what does it replace, and what are its three costs?
- 6 Why is a two-apart spring an inadequate bending model?
- 7 Give the four layers of Bridson's collision treatment and say what the last one guarantees.
- 8 Give three ways hair differs from cloth.
- 9 Why does hair hold its shape, and what does that imply for simulation?